[YOUR COLLEGE NAME]
DEPARTMENT OF [YOUR DEPARTMENT]

System Study

For

AI-Powered Smart Farming Assistant

Submitted in partial fulfillment of the requirements for
9th Semester Mini Project – Review 1

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1 Introduction

This document studies the current (manual) way farming decisions are made in Kerala, identifies its limitations, and presents the proposed system as a response to those limitations. It also evaluates whether the proposed system is realistically achievable within this project's technical, economic, operational, and schedule constraints.

2 Existing System

Currently, farmers in Kerala make most cultivation-related decisions manually, based on experience, informal advice, or trial and error, rather than data-driven analysis.

2.1 How Decisions Are Currently Made

- **Crop selection** is based on tradition, habit, or what neighbouring farms are growing, rather than an analysis of the specific soil and climate conditions of a given plot.
- **Disease identification** depends on the farmer's own experience or a visit from an agricultural officer, which is not always quickly available.
- **Selling decisions** are often made without knowledge of current market prices, leaving farmers dependent on middlemen.
- **Weather-related precautions** rely on general awareness rather than integrated, location-specific alerts.
- **Intercropping** (growing multiple crops together, common in Kerala homesteads) is practiced informally, based on inherited knowledge, without structured guidance on which combinations are most productive.

2.2 Limitations of the Existing Approach

Limitation	**Impact**
No data-driven crop guidance	Inconsistent yields; land may be used for a less-suitable crop than the soil/climate would otherwise support.
Delayed disease detection	Crop damage often spreads before a diagnosis is available, increasing loss.
No price visibility	Farmers frequently sell below fair market value due to dependence on middlemen.
No integrated weather warning	Crops can be damaged by unexpected rain or heat with no advance notice.

Limitation	Impact
Unstructured intercropping	Potential additional income from the same land is left unrealized due to lack of structured guidance.
Fragmented information	Weather, price, and crop-health information each require separate, disconnected sources – if consulted at all.

3 Proposed System

The proposed system, the AI-Powered Smart Farming Assistant, addresses the above limitations through a single mobile-first application that brings together crop recommendation, disease detection, weather intelligence, and market intelligence, with Kerala as its primary geographic focus.

3.1 Overview

The system allows a farmer to:

- Enter basic soil values and get an ML-based crop recommendation, along with a Kerala-relevant intercropping suggestion.
- Capture or upload a photo of a diseased leaf and receive an automated diagnosis with treatment guidance.
- View localized weather forecasts and receive alerts before damaging conditions occur.
- Check current Kerala mandi prices for their crop, with the option to compare against neighbouring states.
- Receive proactive notifications instead of having to actively check multiple sources.

3.2 Advantages Over the Existing System

Existing System	Proposed System
Crop choice based on habit/guesswork	Crop choice based on a trained ML model using actual soil and climate inputs
Disease identified late, informally	Disease identified early, from a photo, in seconds
No visibility into market prices	Live Kerala mandi prices with cross-state comparison
Informal intercropping knowledge	Structured intercropping suggestions based on real Kerala homestead practices

Existing System	Proposed System
Reactive to weather	Proactive weather alerts before damage occurs
Information scattered across sources	One integrated mobile application

3.3 Objectives

1. To recommend suitable crops based on soil and climate data specific to Kerala.
2. To suggest scientifically and traditionally sound intercropping combinations.
3. To detect crop diseases from images and suggest treatment.
4. To provide accessible market price information to reduce dependency on middlemen.
5. To deliver timely weather-based alerts.
6. To package all of the above into a single, simple, mobile-first application usable by farmers with limited digital experience.

4 Feasibility Study

4.1 Technical Feasibility

All required technology is free-tier and well-documented: React Native/Expo for the mobile app, FastAPI for the backend, Scikit-learn/TensorFlow for the ML models, and free APIs (weather, government market-price data) for external data. The team has the necessary skill overlap (ML, backend, frontend) across its three members. **Verdict: Feasible.**

4.2 Economic Feasibility

The system is built entirely on free-tier tools, free datasets, and free-tier hosting, so there is no direct monetary cost for development or for running the semester-scale demo. **Verdict: Feasible, at zero direct cost.**

4.3 Operational Feasibility

The target users (Kerala farmers) already use smartphones for daily communication, so the underlying device requirement is realistic. The interface is designed to require minimal digital literacy (a few taps per task). Manual soil-value entry is used instead of IoT sensors in the current scope, which keeps operational dependencies low. **Verdict: Feasible, with the interface kept deliberately simple.**

4.4 Schedule Feasibility

The project is planned across a full 16-week semester with three team members, split across ML, backend/data, and frontend responsibilities, with review checkpoints built into the timeline (see the Implementation Plan document). **Verdict: Feasible, provided the phase-wise plan is followed.**

4.5 Data Feasibility (Project-Specific Risk)

Since this system depends heavily on real data, dataset availability was separately assessed per feature:

- Crop recommendation and yield prediction: usable public datasets exist, with some Kerala-specific crop gaps that require supplementing.
- Disease detection: strong public datasets exist for common crops; Kerala plantation crops (coconut, pepper, rubber) require self-collected images.
- Market data: a real government data source exists (Agmarknet/data.gov.in), though Kerala-specific listings are comparatively sparse.

This risk is acknowledged and mitigated in the project's build order (self-collection budgeted early, cross-state comparison as a fallback for thin market data).

5 Scope of the Study

This system study, and the project as a whole at this stage, covers the Smart Farming module only, focused on Kerala, across four feature domains: AI/ML crop and yield prediction (with intercropping as an innovation focus), basic soil intelligence, disease and pest management, and market intelligence (with cross-state comparison as an innovation focus). IoT hardware integration and the Smart Gardening module are not part of the current implementation scope.

6 Summary

The existing, manual approach to farming decisions in Kerala leaves farmers without timely, data-backed guidance on crop choice, disease response, or fair pricing. The proposed AI-Powered Smart Farming Assistant addresses each of these gaps directly, and has been assessed as technically, economically, operationally, and schedule-feasible for completion within the current academic semester, with known data-availability risks identified and planned for.